

Criticality and Chiral Boundary Modes in Dynamical Chern Insulators

Ultimate Quantum Matter¹

¹*Department of Physics, Cornell University, Ithaca, NY 14853, USA*

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One of the defining features of topological phases of matter is the bulk-boundary correspondence: gapless modes emerge at the interface between phases with distinct topological invariants, with their existence mandated by the bulk topology rather than by any local order parameter [1–3]. In this work, we study non-equilibrium analogues of these modes in quantum circuits driven by projective measurements and unitary feedforward, also known as *adaptive circuits*. Although measurements are stochastic and non-unitary, free-fermion circuits with measurements admit a complete, symmetry-based topological classification via a generalization of the Altland–Zirnbauer (AZ) tenfold way [4–7]. In this picture, the steady-state quantum trajectories reside in an ensemble of area-law entangled states sharing the same topological invariant. While the bulk classification is now well established, the boundary mode physics in genuinely monitored settings has received limited attention.

The closest prior work is by Xiao and Kawabata [5], who showed that imposing open spatial boundaries on a topological dynamical system produces gapless modes in the Lyapunov spectrum of the stochastic transfer matrix. Their analysis is a classification result for the transfer-matrix dynamics rather than a state-preparation protocol: realizing the boundary mode in physical observables of a specific prepared state requires either post-selection on rare measurement outcomes or additional structure to ensure the trajectory ends in the desired phase. Post-selection alone is exponentially costly in system volume [8, 9], and moreover trivializes the preparation problem entirely: by conditioning on favorable measurement outcomes, one can effectively implement imaginary-time cooling $e^{-\beta H}$ toward any target ground state.

Other recent work has extended the boundary analysis but only within one spatial dimension: Oshima *et al.* [10] characterized topological sectors of the Lyapunov spectrum at measurement-induced transitions in 1D, and Pan, and Jian [11] demonstrated Majorana zero modes in genuinely monitored 1D circuits without post-selection.

Finally, [4] presented some phenomenological work on topological boundary mode physics at domain walls of a 2+1D adaptive circuit, but the understanding of the physics was limited. More precise answers to questions about chirality, topological protection, slow relaxation, and criticality are unclear in the genuinely monitored setting for these 1D dynamical boundary modes.

Other programs have approached non-equilibrium

topology through Floquet engineering or Lindbladian dissipation, with limitations of their own. Floquet topological phases can host chiral edge modes in their quasienergy spectrum [12–15], but the framework requires discrete time-translation symmetry that is broken by the inherent temporal disorder of random measurement sequences.

On the other hand, Lindbladian preparation of topological states have also been studied extensively [16–18]. Lindbladian evolution does not guarantee a pure steady-state and represents continuous-time evolution of the density matrix at the trajectory-averaged level. Indeed, any trajectory-averaged adaptive circuit has a one-to-one mapping with an associated Lindbladian [4, 19]. Because of this, the presence of domain walls leads to mixed steady-states. Furthermore, Goldstein’s no-go theorem [20] states that no finite-range Lindbladian can drive finite-rate exponential relaxation toward a unique pure topological state in $d \geq 2$; pure-state preparation requires exponentially-local dynamics, while finite-range Lindbladians can only prepare mixed states arbitrarily close to the target. This means that dissipative preparation inherently requires jump operators that have support over the whole lattice i.e. have non-finite support.

In this work, we construct an adaptive circuit in $2 + 1$ dimensions that uses local measurements and unitary feedforward to steer toward a pure state ensemble of disordered Chern insulators with boundaries. The adaptive circuit consists of a periodic schedule of truncated, finite-support coherent-state occupancy measurements and unitary feedforward with a bath to correct undesired measurement outcomes. This is conceptually similar to a dynamical quantum code running active error correction. However, the finite-support of the coherent-state measurements introduce spatial disorder, generating an ensemble of pure Gaussian steady-states, each carrying a definite Chern number $C = 1$.

We identify and study two distinct ways to prepare a spatial topological domain wall adaptively; domain walls can be generated either by measuring coherent-state occupancies of distinct topologies restricted to a partition of the lattice, or equivalently, by truncating the coherent-state modes at the boundary of a uniformly topological region. These protocols steer towards a charge-fluctuating ensemble of quasi-1D critical states. Indeed, we observe critical entanglement scaling stemming from the boundary, matching that of the Cardy-Calabrese formula with central charge $c = 1$, while the bulk contributes area-law entanglement. Furthermore, we show

that the dynamics of the perturbation at the boundary are localized by studying the Lyapunov spectrum of the dynamics. Indeed, the single-particle Lyapunov spectrum becomes gapless in the thermodynamic limit when a domain wall is present, and the corresponding Lyapunov eigenvector is sharply localized at the domain-walls. Finally, we confirm power-law scaling of the correlation function along the boundary.

Our work is perhaps the first post-selection free, finite-support protocol that can prepare chiral cft physics at the boundaries of topologically robust regions.

Constructing a dynamical Chern insulator. We wish to build a quantum circuit in $2+1$ D that uses local measurements and feedforward to steer toward a disordered ensemble of Chern insulators.

To motivate this construction, let us first consider the parent Hamiltonian for a Chern insulator in equilibrium [AB: (continue here with the model)].

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